

A Load-Cell Cup Detection Module for the RoboCoffee Delivery Fleet

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Abstract—This paper presents a cup detection module for the RoboCoffee intralogistics fleet. The module measures the load on the cup holder with a strain-gauge load cell and reports placement and removal events to the central platform. A threshold rule with hysteresis is proposed, and a preliminary test of 20 placement–removal cycles is reported.

Index Terms—Service robots, intralogistics, load cell, object detection, mobile robots.

I. INTRODUCTION

DIFFERENTIAL-DRIVE mobile robots are a mature platform for indoor service tasks, since their kinematics and odometry are well understood [1]. In a coffee-delivery service, however, the robot must also know whether the order is still on board: a delivery is only complete when the client removes the cup.

When several robots share the same pool of orders, the waiting time of each client depends on how quickly a robot is released after each delivery, which can be analyzed with queueing models [2]. A robot that cannot confirm the removal of the cup either waits a fixed time, which wastes fleet capacity, or leaves too early, which risks returning with an undelivered order.

This work addresses the following research question: can a single load cell under the cup holder detect cup placement and removal reliably enough, and fast enough, to release the robot automatically at the end of each delivery?

The contributions of this paper are: (i) a low-cost detection module that uses a load cell and an ESP32 microcontroller; (ii) a threshold rule with hysteresis that rejects the vibration produced while the robot moves; and (iii) a test protocol that quantifies detection reliability and latency.

II. RELATED WORK

Radio-frequency identification (RFID) is a common way to detect tagged objects [3]. However, RFID requires a tag on every cup, which is not compatible with the disposable cups used in a cafeteria. Measuring the weight of the cup does not require any tag and also distinguishes an empty holder from a full or an empty cup.

III. MATERIALS AND METHODS

The architecture of the module is shown in Fig. 1. A 1 kg strain-gauge load cell supports the cup holder; an HX711 amplifier converts its signal at 80 Hz, and an ESP32 applies the detection rule and sends events to the central platform over Wi-Fi.

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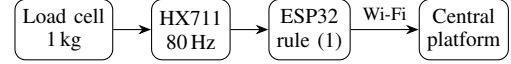


Fig. 1. Architecture of the cup detection module.

TABLE I
TEST PROTOCOL

Cups tested	empty (12 g), full (250 g)
Cycles	20 (this cut), 100 (next cut)
Metrics	detection rate, false events, latency

Let m_k be the filtered mass at sample k . The state s_k of the holder changes only when the mass crosses one of two thresholds:

$$s_k = \begin{cases} 1, & m_k > m_{\text{on}}, \\ 0, & m_k < m_{\text{off}}, \\ s_{k-1}, & \text{otherwise,} \end{cases} \quad (1)$$

with $m_{\text{on}} = 40$ g and $m_{\text{off}} = 20$ g. The gap between both thresholds prevents false events caused by the vibration of the holder while the robot moves.

The test protocol is summarized in Table I. Each cycle consists of placing a cup, moving the robot 3 m, and removing the cup; the latency is the time between the removal and the event received by the platform.

IV. PRELIMINARY RESULTS

In the 20 preliminary cycles with a full cup, all placements and removals were detected and no false events occurred while the robot was moving. The mean removal latency was 0.31 s. These results correspond to a single cup type and a single robot; the complete campaign of 100 cycles with both cup types will be reported in the next cut.

V. DISCUSSION AND LIMITATIONS

An empty disposable cup (12 g) lies below m_{off} , so the current rule cannot distinguish an empty cup from an empty holder. Camera-based detection could identify the cup independently of its weight, but it would add an image-processing pipeline and lighting requirements outside the scope of this module; it is therefore left as future work.

VI. CONCLUSIONS

A load-cell module with a hysteresis rule detected cup placement and removal in all preliminary cycles with a sub-second latency, which supports its use to release RoboCoffee robots automatically after each delivery.

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